

ARTICLE

Excitonic coupling of electronic transitions in oligomeric phthalocyanine zinc(II) complexes

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The excitonic coupling of the electronic transitions in a series of zinc(II) phthalocyanine (ZnPc) dimers has been studied by TD-DFT with a range-separated hybrid functional LC-BLYP. This method correctly reproduces the experimental excitonic shifts of the Q-band for known "rigid" non-conjugated binuclear phthalocyanines. The observed significant differences in the experimental UV/Vis spectra of the "edge-to-edge", "slipped-stacked" and "face-to-face" ZnPc dimers have been explained. The same TD-DFT method has been used to investigate the influence of the relative position of ZnPc molecules on the Q-band transition energies. The Coulombic contribution to excitonic coupling (according to Kasha's model) has been estimated using the transition electrostatic potential (TrESP) atomic charges. The excitonic splitting of the Q-band has been extrapolated from Pc oligomers to an infinite 1D-stack using the tight-binding model, and the main spectral features of α - and β -modifications of ZnPc have been correctly reproduced.

Introduction

Phthalocyanines (Pc) have found many "real" applications as colorants and materials for electronic devices.^{1–3} Recent potential applications include organic thin-film transistors,^{4,5} near-infrared fluorescent diagnostics and photodynamic immunotherapy of tumors.⁶ Successful practical use of phthalocyanines (Pcs) in many fields originates from their structural versatility, photophysical properties and a wide range of solid-state arrangements.

The photophysical properties of phthalocyanine metal complexes can be modified by peripheral substituents, central metal ions or by extension of the π -system.^{7–9} Noncovalent interactions (excitonic coupling) of the Pc π -systems also significantly influence the photophysical properties of phthalocyanines in aggregates, non-conjugated oligomers and crystals.^{10–35}

Non-conjugated phthalocyanine assemblies include well-known π - π stacked aggregates in solution¹⁴ and in the solid state,^{2,3,15,16} covalent^{10,24–31,35} and supramolecular^{13,15,18,19–21,26} oligomers. Two types of chromophore arrangement are usually discussed: face-to-face configurations, often referred as H-type aggregates, and edge-to-edge or slipped-stacked configurations known as J-type aggregates (Fig. 1).¹⁴

According to Kasha's theory, H-aggregates exhibit a blue-shifted absorption maximum and a suppressed radiative decay rate, whereas J-aggregates exhibit a red-shifted absorption

maximum and an enhanced radiative decay rate. For dimers with a parallel arrangement of chromophores, the interaction of the transition dipoles can be approximated by a simple point-dipole model:

$$\Delta E \approx \frac{\mu^2(1-3\cos^2\theta)}{4\pi\epsilon r^3} \quad (1)$$

where μ and r represent the magnitude of the transition moment in the monomer and the centre-to-centre distance between the molecules, respectively, and θ is the angle between the polarization axis and the centre-to-centre line. The energy change ΔE crosses zero at $\theta = 54.7^\circ$, separating H-aggregate ($\theta > 54.7^\circ$) and J-aggregate ($\theta < 54.7^\circ$) behaviours.

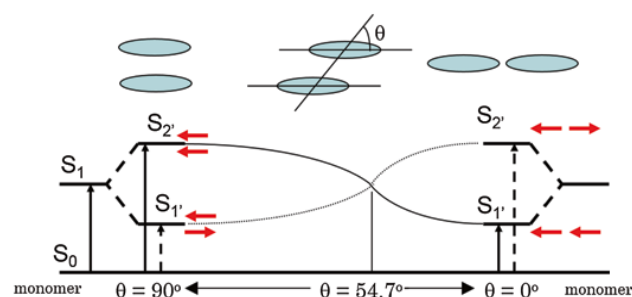


Fig. 1. Excitonic coupling of electronic transitions in molecular dimers.¹² © Hiroaki Isago with CC-BY-NC 3.0 license

Many examples of "face-to-face" phthalocyanine dimers, oligomers and aggregates,^{10,24–28,34,35} and a few "edge-to-edge" and "slipped-stack" dimers and aggregates were described.^{13,18,29–32} But the exact alignment of the Pc subunits in these systems is not known, except for the crystals where slipped-stacked arrangement is typical for M(II)Pcs.^{1,2,16,17} Only a few binuclear Pcs with linkers rigid enough to ensure a well-

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defined geometry are known experimentally (Chart 1): a "gable" **1**,³³ a "planar" **2**,³¹ and a "cyclophane" **3**³² linked binuclear Pcs having "edge-to-edge" and "slipped-stack" arrangements (J-dimers). For the "face-to-face" binuclear Pcs (H-dimers) with rigid linkers such as **4**²⁷ and **5**,²⁸ multiple conformations are still possible, which can influence the resulting excitonic coupling. A supramolecular dimer **6**¹³ can be considered as rigid although two configurations ("parallel" and "oblique") are possible.

In a typical case of H-aggregates, a blue shift from $\lambda = 660\text{--}700$ nm to $\lambda = 600\text{--}640$ nm and a broadening of the Q-band are usually observed.¹⁴ Similar blue shifts were observed for various covalent and supramolecular cofacial dimers.^{10,14,25-28,34,35}

Slipped-stacked Pc oligomers resemble structural patterns of photosynthetic light-harvesting antenna consisting of bacteriochlorophyll dimers,³⁸ and they are photoactive in contrast to face-to-face Pc oligomers^{11-13,39} for which a non-radiative decay of the excited states is typical. The cyclophane-fused dimer **3** is fluorescent ($\Phi_f = 0.12$), a metal-free "gable" dimer **1** is also fluorescent ($\Phi_f = 0.55$), a "planar" dimer **2** shows a singlet oxygen quantum yield of $\Phi_\Delta = 0.08$. The imidazole-bridged supramolecular dimer **6** is also fluorescent ($\Phi_f = 0.26$). Surprisingly, for edge-to edge Pc dimer **2** no redshift predicted by Kasha's theory and eq. (1) was observed. In contrast, J-dimers **3** and **6** exhibit strong bathochromic Q-band shifts (Table 1) in full agreement with Kasha's theory. It was concluded^{8,32} that π -MO overlap is essential for an efficient excitonic coupling in Pc dimers, crystals and aggregates, and purely Coulombic coupling of the transition dipoles (eq. (1)) is not enough to explain the observed Q-band shifts.

In the solid state, β -modification of planar M(II)Pcs is characterized by a close contact between the central metal ion and a meso-N atom of the adjacent molecule, with an inclination angle of $\sim 45^\circ$ within the stack. The inclination angle in the α -modification is $25\text{--}30^\circ$, and the direction of the tilt may vary.^{16,36,37} Different chromophore arrangements result in a significantly different excitonic coupling, leading to a high

variability of absorption wavelengths of crystalline phthalocyanines (Fig. 2). Thus, α -modifications of CuPc and ZnPc absorb at $\lambda \sim 625$ and ~ 700 nm, β -modifications at $\lambda \sim 650$ and ~ 750 nm, and ϵ -CuPc exhibits an absorption at even longer wavelength of $\lambda < 820$ nm.¹⁻³ The non-planar Ti(O)Pc with slipped-stacked arrangement similar to J-aggregates shows a strongly red-shifted Q-band absorption at $\lambda \sim 700\text{--}900$ nm and an efficient photoinduced charge carrier generation.³³

The excitonic coupling in phthalocyanine dimers³⁹⁻⁴¹ and crystals³⁹⁻⁴² was previously studied by DFT and semi-empirical methods. It was demonstrated that short-range coupling is very sensitive to MO overlap^{39,41} and is related to charge-transfer integrals.^{41,43} This explains the pronounced variability in solid-state UV/Vis spectra among structurally isomorphic Zn(II), Ni(II) and Cu(II) Pcs.⁴⁰ For selected arrangements, short-range excitonic coupling in ZnPc molecular dimers has been also investigated by TD-DFT.⁴⁴ However, a systematic study of dimer arrangements (r and θ) is still required to provide an accurate description of excitonic coupling and other intermolecular properties, forming a baseline for emerging machine-learning models geared toward crystal structure prediction.⁴⁵

Previously, we demonstrated that the range-separated hybrid functional LC-BLYP accurately reproduces the UV/Vis/NIR spectra of conjugated ZnPc oligomers.^{8,9} The extrapolated TD-DFT Q-band energy of 1D-oligoPcs is comparable to a directly calculated band gap of 1D-polyPc.

Here, we employ the same LC-TDDFT protocol to investigate systematically the dependence of the Q-band excitonic splitting on the relative spatial arrangement (r and θ) of ZnPc chromophores, and finally to explain the excitonic effects observed in phthalocyanine aggregates, dimers and crystals. The applicability of the above-mentioned TD-DFT protocol is validated against a benchmark set of experimentally known, structurally rigid non-conjugated binuclear ZnPcs.

Computational details

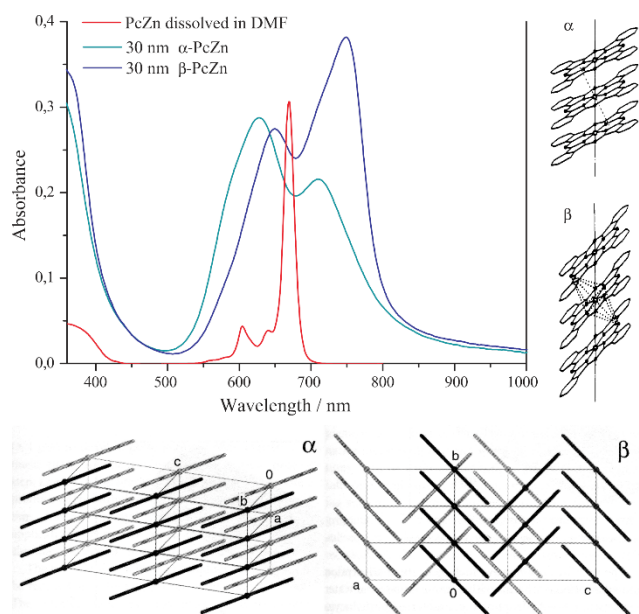


Fig. 2. Electronic spectra and crystal packing of α - and β -modifications of ZnPc

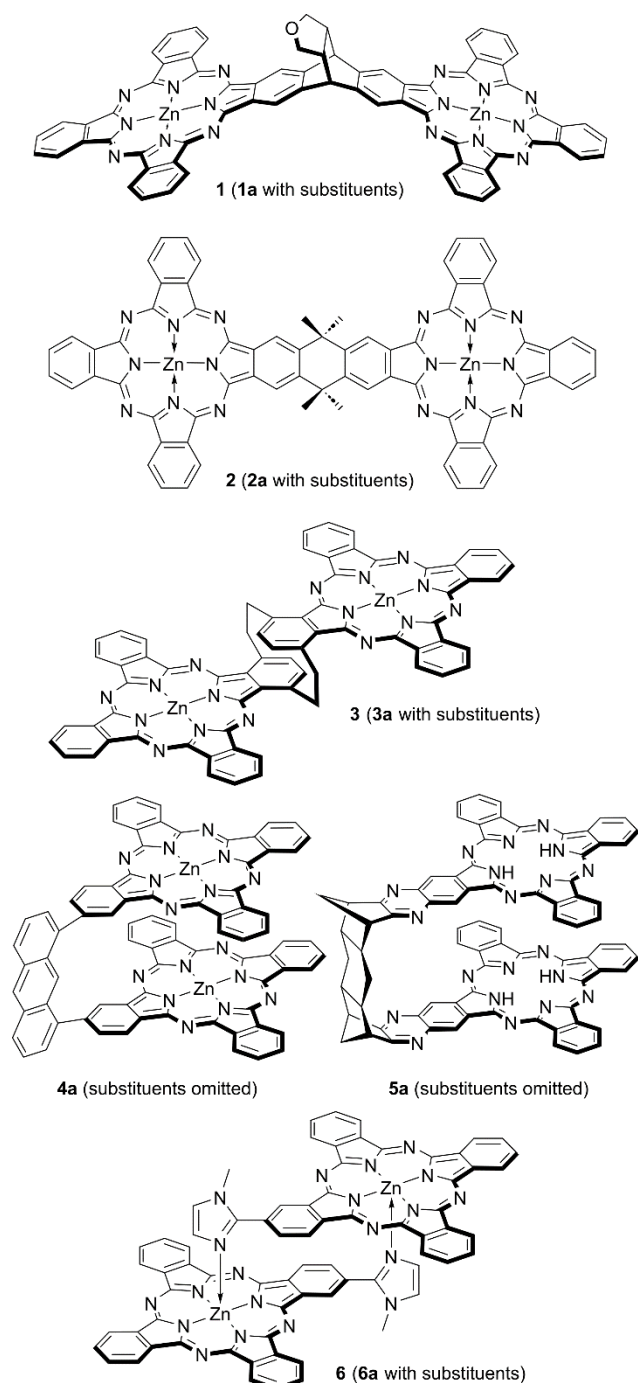


Chart 1. "Rigid" phthalocyanine dimers

DFT calculations. The range-separated hybrid functional LC-BLYP ($\mu = 0.35$)⁴⁶⁻⁴⁸ with the def2-TZVP(-f)/def2-J⁴⁹⁻⁵¹ basis sets were used for TD-DFT calculations, similar to our previous publication.⁸ All geometries were optimised using the r2SCAN-3c functional⁵² which includes a built-in def2-mTZVP basis set, D4 empirical dispersion,⁵³ geometrical counterpoise⁵⁴ and relativistic corrections. The optimization method was chosen from the comparison of the optimised geometries of ZnPc with its experimental gas-phase geometry.⁵⁵ All calculations were carried out in the ORCA 6.0.1 software,⁵⁶ with the RIJCOSX approximation,^{57,58} default grid "Defgrid2",⁵⁶ tight ($\Delta E < 10^{-8}$ a.u.) and very tight ($\Delta E < 10^{-9}$ a.u.) SCF convergence for TD-DFT

and geometry optimizations, respectively. Adding diffuse functions or counterpoise correction had no noticeable effect on the transition energy ($\Delta E < 0.01$ eV).

Transition charges were calculated by fitting the Q-band transition electrostatic potential (TrESP)⁵⁹ of the r2SCAN-3c optimised ZnPc using the CHELPG scheme⁶⁰ as implemented in the Multiwfn 3.8 software.^{61,62}

Extrapolation of TD-DFT spectra was carried out for ZnPc stacks up to the size of 5, constructed from the r2SCAN-3c optimised ZnPc molecules positioned either according to the translation vectors from β -ZnPc⁶³ stacks or by applying shifts of $\Delta x = 3.3$ – 3.4 Å, $\Delta z = 1.75$ – 2.0 Å (Fig. 4A) to simulate the α -ZnPc stacks.

Results and discussion

Bridged binuclear phthalocyanines

"Rigid" binuclear ZnPcs **1**–**3** and **6** were selected to exclude the influence of noncovalent interactions (poorly described by DFT⁶⁴) on the relative position of the Pc chromophores.

The calculated Q-band wavelengths and energies for the dimers **1**–**3**, **6** without peripheral substituents were compared to experimental wavelengths of their Q-band maxima. The results are listed in Table 1.

Table 1. Experimental and calculated Q-band wavelengths, energies and intensities for rigid non-conjugated binuclear ZnPcs **1**–**3**, **6** and their substituted analogues **1a**–**3a**, **6a**

Cpd.	Experimental (1a – 3a, 6a)		Theoretical (1 – 3, 6)	
	λ , nm (ΔE , eV)	$10^{-5} \epsilon$, $M^{-1} \cdot cm^{-1}$	λ , nm (ΔE , eV)	Oscillator strength
ZnPc ⁸	674 (1.840)	3.4	673 (1.842)	0.472 · 2
1 ,	694 (1.787)	2.9	699 (1.773)	1.054
1a ³¹	677 (1.832)	3.1	669 (1.854)	0.885
			662 (1.871)	0.194
2 ,	682 (1.818)	6.9	693 (1.790)	1.313
2a ²⁹			681 (1.821)	0.889
3 ,	753 (1.647)	0.84	742 (1.670)	1.262
3a ³⁰	690 (1.797)	0.72	679 (1.826)	0.794
6 ,	700 (1.771)	1.7	715 (1.733)	1.070
6a ¹²	670 (1.851)	1.2	669 (1.852)	0.749

As can be seen from Table 1, the theoretical wavelengths are in agreement with the experimental ones. Therefore TD-DFT with a range-separated hybrid functional LC-BLYP correctly predicts the "through-space" excitonic coupling of the excited states of ZnPcs. For the edge-to-edge arrangements in **1** and **2**, a redshift can be expected from eq. (1). However, the observed and predicted redshifts are negligible. This contrasts with a well-

known blue shift for face-to-face arrangements in H-dimers and aggregates.^{10-13,25-28,34,35} Conversely, the expected redshifts are evident for "slipped-stack" arrangements in **3** and **6**. Especially for the cyclophane-bridged dimer **3**, the red shift is substantial and even comparable to some conjugated Pc dimers.⁷ The nearly zero red shift for the edge-to-edge dimers **1** and **2** was explained^{8,30} by the absence of the π - π overlap. Thus, it is important to study whether a general Kasha's dipole-dipole model is still applicable, or the observed excitonic coupling in phthalocyanine aggregates, dimers and crystals is a system-specific short-range effect determined by the π - π overlap.

Excitonic coupling between phthalocyanine molecules

To evaluate excitonic coupling effects in oligoPcs, model zinc phthalocyanine dimers with face-to-face, slipped-stack and edge-to-edge arrangements of the Pc rings with parallel shifts (Δx) from 0 to 20 Å and interplane distances (Δz) from 0 to 6 Å were studied by TD-DFT. The corresponding distances (r) and angles (θ) can be easily calculated to compare TD-DFT transition energies to those predicted by equation (1). Phthalocyanines show intense $\pi \rightarrow \pi^*$ absorption bands in the visible region of their absorption spectra. The Q-band corresponds to the doubly degenerate HOMO (a_{1u}) $\rightarrow$ LUMO (e_g) transition (Fig. 3).⁶⁵ In a Pc dimer, four Q-band transitions arise from the excitonic splitting of two degenerate E_u transitions of a single Pc chromophore. Two allowed B_u and A_u transitions ($f \approx 0.5$) are x- and y-polarised, respectively. The remaining A_g and B_g transitions are forbidden ($f \approx 0$) due to the opposite orientation of the transition dipoles. Unlike nearly "pure" Q-band

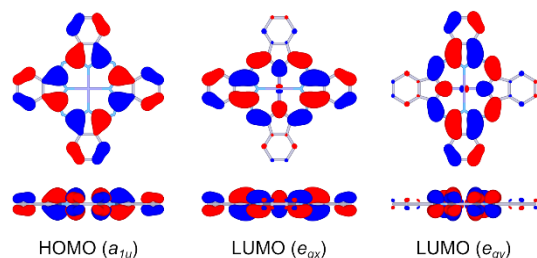


Fig. 3. Frontier MOs of ZnPc (front and side views)

transitions of monomolecular Pcs, the configuration of each Q-band transition of a Pc dimer is a combination of two main transitions involving frontier orbitals (Fig. 5). The MOs of the Pc oligomers can be approximated as linear combinations of the monomer orbitals.¹¹⁻¹³ Due to π - π overlap, bonding-antibonding interactions between these MOs result in splitting of the corresponding energy levels. In infinite stacks, this results in the formation of valence and conduction bands.^{39,43} The dependence of these energies on the arrangement of the Pc molecules is shown in Fig. 6 for the shift direction along Zn–N bond (Fig. 4A). The corresponding transition energies are listed in Table S1 of the ESI[†]. For the shift direction along *meso*-N – Zn (Fig. 4B, which approximately resembles the stack arrangement in β -ZnPc), the results are shown in Fig. S1 and Table S2 of the ESI[†].

Angle dependence. For parallel transition dipoles, the dependence of the excitonic splitting on the angle θ (Fig. 1) is approximately described by equation (1). With "face-to-face" arrangements, the allowed B_u (Q_x) transition is blue-shifted in contrast to the corresponding forbidden A_g transition, which is

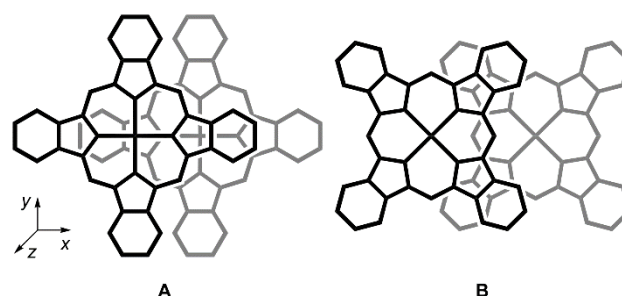


Fig. 4. Arrangements of the monomers in slipped-stacked ZnPc dimers

strongly red-shifted and can participate in non-radiative decay

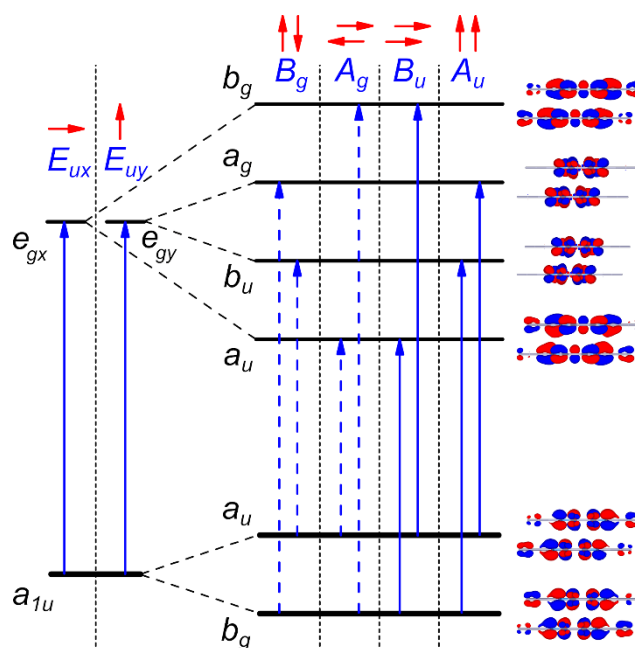


Fig. 5. Qualitative scheme of frontier MO splitting and Q-band transitions of slipped-stacked ZnPc dimers.

of the excitation energy. This explains why "face-to-face" dimers are non-fluorescent. For the "slipped-stack" configurations with larger shifts (Δx), the allowed B_u transition becomes red-shifted with increasing dislocation and its energy drops below the corresponding A_g forbidden transition. This explains why the "slipped-stack" and "edge-to-edge" dimers are fluorescent. The angle corresponding to a zero splitting between B_u and A_g transitions is 54.7° , which is indicated by a vertical line in Fig. 6.

As can be seen from Fig. 6, there is a qualitative agreement between the TD-DFT results and equation (1). At the 3.5 Å interplane distance, the dependence of energy splitting on the shift is irregular due to short-range effects of the MO overlap (see below). These effects are much less pronounced already at the 4.0 Å distance. For the edge-to-edge arrangement ($\Delta z = 0$, see Fig. S2 of the ESI[†]), only large centre-to-centre distances $r = \Delta x \geq 16$ Å are physically possible, which results in very small excitonic splittings, in agreement with eq. (1). For the "planar" dimer **2**, this distance is 15.7 Å, and the predicted excitonic splitting is only 0.03 eV. Thus, the small redshift for the edge-to-edge dimers **1** and **2** is simply explained by the large distance between the centres of the transition dipoles. However, there is no quantitative agreement between the TD-DFT transition energies and eq. (1), which can be explained by short-range effects of the π -MO overlap,^{39–44} as discussed below.

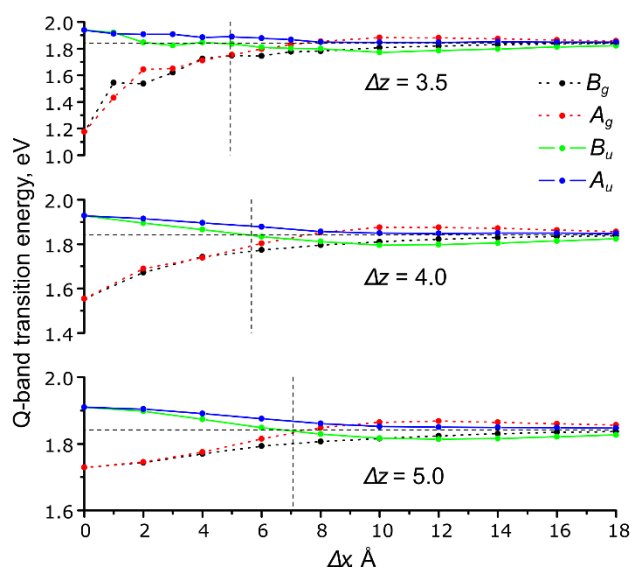


Fig. 6. TD-DFT Q-band transition energies vs. parallel shifts (Δx) of two Pc molecules (arrangement in Fig. 4A). Forbidden transitions are shown by dashed lines. The horizontal line represents the Q-band energy of ZnPc, the vertical line corresponds to the zero-splitting angle of 54.7°

Distance dependence. As shown by Harcourt *et al.*,⁶⁶ the excitonic interactions can be divided into long-range Coulombic interactions, Dexter's short-range exchange,⁶⁷ and superexchange caused by the π -MO overlap.⁴³ Taking into account only the Coulombic interactions leads to Kasha's exciton model²³ often approximated as a point-dipole interaction (eq. 1). This scales as $1/r^3$ whereas the superexchange interaction depends on the MO overlap, which decays exponentially with distance.

To exclude the influence of the variable π -MO overlap, only perfect face-to-face ($\Delta x = 0$) and edge-to-edge ($\Delta z = 0$) arrangements were considered in order to study the distance dependence of the excitonic Q-band splitting. As can be seen from Fig. 7, at larger Δz distances the splitting dependence is almost linear with respect to $1/r$ ($r = \Delta z$) whereas the point-dipole approximation (eq. (1)) predicts $1/r^3$ decay. The Coulombic component of the excitonic coupling was evaluated using the atomic charges obtained from the fitting of the transition electrostatic potential (TrESP charges).⁵⁹

At larger distances, the TD-DFT transition energies for the dimers are well reproduced by the TrESP Coulombic contribution calculated from monomers' transition charges. Within the studied distance range, the Coulombic component scales linearly with $1/r$ rather than $1/r^3$: the distances between the transition dipoles are still comparable to the size of the macrocycles, therefore the point-dipole approximation is not yet valid. Significant disagreement between the TD-DFT transition energy and the predicted Coulombic component (Fig. 7) at shorter interplane distances (Δz) indicates a large impact of the short-range effects on the excitonic coupling. Another noticeable effect is a non-symmetrical splitting at shorter distances: the red shift of the A_g transition is larger than the blue shift of the B_u transition. This can be explained by a larger contribution of a lower-energy transition to the configuration of the excited state (Table S3 of the ESI[†]). The red-shifted A_g transition is forbidden, therefore H-aggregates show only a blue-shifted Q-band. In crystals, the red-shifted Q-band absorption component often extends beyond $\lambda = 800$ nm (shifts of $\Delta E > 0.3$ eV) whereas the blue-shifted band is usually observed at $\lambda = 600 - 650$ nm (shifts of $\Delta E = 0.1 - 0.2$ eV). For the edge-to-edge arrangement ($\Delta z = 0$), the TD-DFT Q-band splitting is only 0.05 eV at $\Delta x = 17$ Å and is purely Coulombic. Therefore, the moderate blue shift ($\Delta E \sim 0.15$ eV) typically observed for Pc H-aggregates,^{10,14,25–28,34,35,65} is merely the "tip of the iceberg": if the A_g transition were allowed, it would appear at $\lambda > 1000$ nm. Thus, the short-range intermolecular electronic interactions in the Pc H-aggregates are much stronger than expected from the experimental UV/Vis spectra. However, this strong electronic coupling can be reflected in other properties. Specifically, it contributes to the excited-state quenching in aggregated Pcs^{11–14} (since the forbidden excited state is the lowest one) and correlates with the efficient charge transport in Pc-based molecular conductors^{17,68} (as short-range excitonic coupling is related to charge-transfer integrals^{11,43,68}).

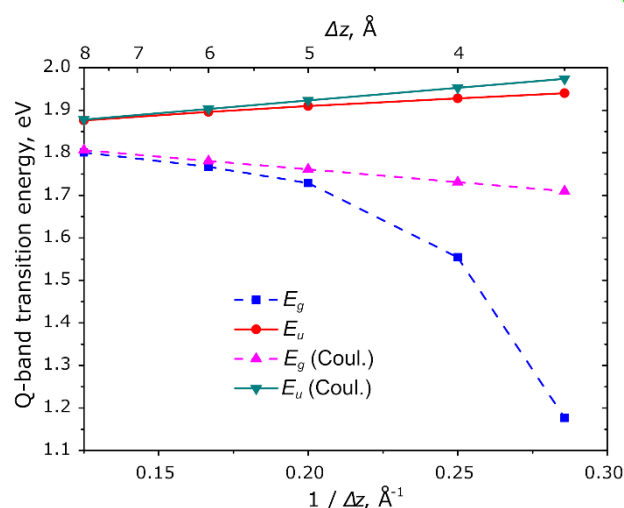


Fig. 7. Distance dependence of the TD-DFT Q-band transition energies of a face-to-face ZnPc dimer

MO analysis. The π - π MO overlap results in splitting of the HOMO and LUMO levels of the monomers as shown in Fig. 8. The HOMO and LUMO splitting pattern in the slipped-stack

dimers is determined by the MO shapes of the corresponding monomers.^{39,43} The MO overlap is responsible for short-range effects (see above) often expressed as charge-resonance/charge-transfer (CR/CT) excitons.^{11,43} MO overlap and, as a consequence, the charge-transfer integrals^{43,59} are very sensitive to intermolecular arrangement. This results in a great variability of photophysical and electric properties of crystalline modifications of phthalocyanines.^{1-3,16,70} A parallel shift between Pc molecules in the dimer results in oscillations of the split MO levels with different periodicity for HOMO and LUMO which is a result of different nodal patterns of the monomers' HOMO and LUMO (Figs. 4, 8, Table S4).³⁹ These MO level oscillations result in irregular shift dependence of the corresponding transition energies (see Fig. 6). The splitting generally decreases with the shift due to decreasing overall π -MO overlap. The dependence of the splitting on the distance is exponential as a result of the exponential decay of the π -MO overlap (Fig. S3 of the ESI[†]). This is in contrast to the $1/r$ dependence of the TD-DFT transition energies at larger distances. Therefore, at the distances of 5–8 Å the MO splitting becomes negligible but the TD-DFT transition energies still show

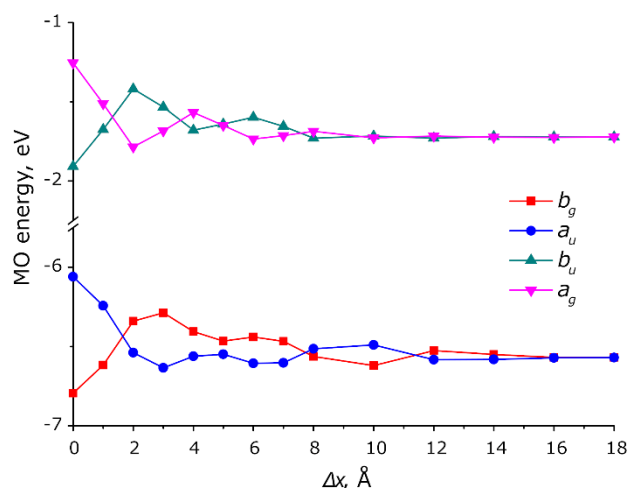


Fig. 8. Energies of 2 highest occupied (a_u , b_g) and 2 of 4 lowest vacant (a_g , b_u) orbitals of the ZnPc dimers at $\Delta z = 3.5$ Å and various shift values (Δx)

a non-zero excitonic splitting as a result of Coulombic coupling of the monomers' transition dipoles.

Size dependence. Previously, the optical gaps of π -conjugated Zn(II) oligoPcs were extrapolated to a 1D-polyPc using a free-electron approximation⁸ (E vs. $1/N$), which is applicable only to delocalised systems.⁷¹ However, for typical molecular crystals, orbital and optical gaps can differ significantly.⁷² Therefore, the extrapolation of TD-DFT optical gaps of oligomers may provide a better estimate for the optical gap of an infinite stack than the orbital gap directly calculated under periodic boundary conditions.

According to a simplified tight-binding model including only the nearest-neighbour interactions between ZnPc molecules,^{39,40} the band dispersion (width of HOMO or LUMO band) for an N -meric stack is described by the following equation (2).

$$\Delta E = \pm 2J \cos[\pi / (N + 1)] \quad (2)$$

The coupling factor J is either J_{Coul} or J_{CT} for Coulombic or charge-transfer excitonic coupling, respectively. As it was shown by Bredas *et al.*, the HOMO and LUMO band edges are also linear with respect to $\cos[\pi/(N+1)]$ for phthalocyanines and other molecular semiconductors.³⁹ Similar extrapolation was also applied for optical gaps of oligothiophenes.⁷³

The excitonic splitting of two Q-band transitions results in $2 \cdot N$ transitions for the N -meric ZnPc. The TD-DFT spectra of ZnPc oligomers are dominated by two x - and y -oriented transitions (A_{ux} and A_{uy} , respectively, see Fig. 3) which correspond to all monomers' transition dipoles (Fig. 5) pointing in the same

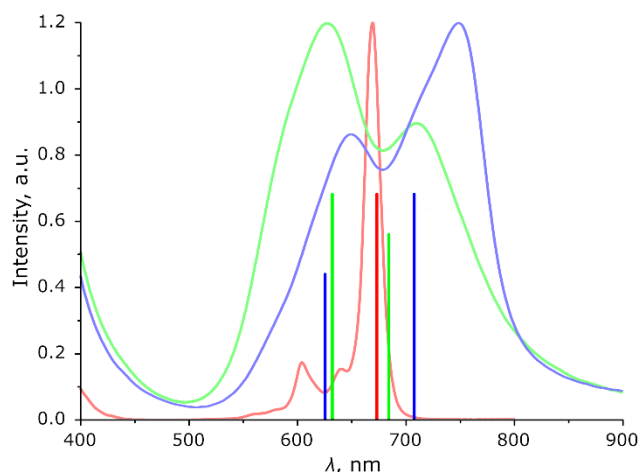


Fig. 9. Main TD-DFT Q-band transitions (vertical lines) of ZnPc (red), ZnPc oligomers extrapolated to infinite stacks approximately reproducing the α - (green) and β - (blue) ZnPc stacks, together with experimental spectra of ZnPc in DMF, α - and β -ZnPc films, presented in the corresponding colours

direction. All other Q-band transitions are of zero or negligible intensity ($f < 0.02$).

Equation (2) can be used to predict the excitonic splitting of the optical spectra for infinite 1D-stacks in aggregates or crystals.

To study the size dependence of the excitonic splitting, larger aggregates (stacks) were calculated with the arrangements approximately resembling the β -modification of ZnPc (translation vectors taken from its X-Ray structure⁶³). As no single-crystal X-Ray structure is available for α -ZnPc, estimated model stacks were studied. These include structures with a Δz of 3.3–3.4 Å and a Δx of 1.75–2.0 Å in the direction along the Zn–N bond (Fig. 4A), resembling the redetermined “brickstone” α -CuPc structure,³⁶ as well as along *meso*-N – Zn direction (Fig. 4B), similar to the classical “herringbone” α -CuPc structure¹⁶), were studied. The resulting Zn–Zn distance is 3.75–3.95 Å with a tilt angle of 28–31°, which is comparable to previous estimates.^{40,42} The results are shown in Fig. S4 and Table S5 of the ESI[†]. As can be seen from Fig. S4, the TD-DFT Q-band splitting is almost linear with respect to $\cos[\pi/(N+1)]$ for $N = 1...5$ which is in agreement with eq. (2). This indicates that the excitonic splitting of the Q-band is mostly determined by the nearest-neighbour interactions. The extrapolated to $N = \infty$ transition energies of 1.98 eV ($\lambda = 626$ nm) and 1.75 eV ($\lambda = 708$ nm) for a single stack of β -modification are comparable with the experimental values (Fig. 2) of 1.94 eV (640 nm) and 1.65 eV (750 nm). Oscillator strengths increase almost linearly with size (Fig. S3 of the ESI[†]) and with the extrapolated ratio of 1:1.55

which also agrees well with the experimental spectrum (Figs. 2, 9). Our estimates of the Q-band splitting for α -ZnPc ("brickstone" model, Fig. 4A) vary from 0.08 eV to 0.18 eV (Fig. 9 for $\Delta x = 2.0$ Å and $\Delta z = 3.4$ Å, Table S5 of the ESI[†]), *i.e.* very small variations of the intermolecular arrangement result in significant changes of the Q-band splitting. This explains why Zn(II), Ni(II) and Cu(II) Pcs have different solid-state spectra despite very similar crystal structures and solution spectra.⁴¹ For the "herringbone" model stack (Fig. 4B), only blue shift with no splitting is predicted, which is more typical for the aggregated Pcs in solution.^{10,14,25-28,34,35} The TD-DFT results within the "brickstone" model³⁶ (Fig. 9) are in a better agreement with the experimental spectrum of α -ZnPc, qualitatively reproducing its main features. The approximate agreement between the extrapolated and the experimental spectra of β -ZnPc confirms that crystal spectra are mostly determined by excitonic coupling within the stack. For a more quantitative description, the next-nearest-neighbour and the inter-stack couplings should also be considered.^{41,74}

Conclusions

Excitonic coupling of the electronic transitions in a series of zinc(II) phthalocyanine dimers was studied by TD-DFT with a range-separated hybrid functional LC-BLYP. This method correctly reproduces the experimental excitonic shifts of the Q-band for known "rigid" non-conjugated binuclear phthalocyanines.

Small excitonic redshift of the Q-band for the "edge-to-edge" Pc dimers is a result of a large distance between transition dipole centres (weak Coulombic coupling) and absence of the π -MO overlap (no short-range coupling). In contrast, the slipped-stacked arrangement of the Pc rings in dimers and crystals results in a strong redshift of the Q-band due to the π -MO overlap and the short-range excitonic coupling. The Kasha's point-dipole approximation (eq. (1)) is qualitatively applicable although the $1/r^3$ dependence is actually a superposition of the $1/r$ dependence for Coulombic coupling and the e^{-r} dependence for short-range coupling. The short-range excitonic coupling is highly sensitive to the intermolecular arrangement due to a complicated HOMO-LUMO nodal pattern which determines the bonding/antibonding character of the π -MO overlap.

The well-known ~50 nm (~0.15 eV) blue shift of the Q-band of the "face-to-face" Pc dimers (H-aggregates) is only a "tip of an iceberg": the red-shifted (by ~0.65 eV) forbidden subband is not observed in the optical spectra but such a strong excitonic coupling indirectly influences other properties. The excitonic splitting of the Q-band can be extrapolated from Pc oligomers to an infinite 1D-stack, in the frame of the tight-binding approximation (eq. (2)). This approach can be further extended to include the next-nearest-neighbour and the inter-stack couplings for predicting optical spectra of crystalline phthalocyanines.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data that support the findings of this study are available in the ESI[†] of this article.

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